

SFT Particle Reality Inventory

The Standard Model Sorted by Persistence:
What Builds the Universe, What Visits Briefly,
and What Was Never There

Reference document for the V13 program

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The universe is assembled from twelve fermions and the photon. Everything else in the Standard Model's particle zoo is either a brief moment during an interaction or a mathematical tool for computing the interaction's outcome.

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1 Three tiers of reality

The Standard Model contains 61+ named particles. Most physics presentations treat them equally — “here are all the particles.” SFT does not. SFT asks: which ones are PERSISTENT self-maintaining structures? Which ones are TRANSIENT settlement events? Which ones are MATHEMATICAL INFRASTRUCTURE that were never physically real?

The answer organizes the entire particle zoo into three tiers, and the universe is built from only the first.

2 Tier 1 — Persistent: the builders

These particles are stable or effectively stable. They exist indefinitely (or for cosmological timescales). They constitute all ordinary matter. They are the closed and open energy orbits that SFT describes as self-maintaining structures.

In SFT: Tier 1 particles are the modes that the theory **MUST** explain. Their masses, charges, spins, and interaction properties are the primary targets.

Particle	Mass	Charge	Spin	Life	Role in the universe
<i>Charged leptons</i>					
tier1Electron	0.511 MeV	−1	$\frac{1}{2}$	Stable	Atomic shells. Chemistry. Biology. Everything you can touch.
<i>Neutrinos</i>					
tier1 ν_e	< 0.1 eV	0	$\frac{1}{2}$	Stable	Produced in beta decay, fusion. Pervades universe. Oscillates.
tier1 ν_μ	< 0.1 eV	0	$\frac{1}{2}$	Stable	Same as above. Distinct flavor.
tier1 ν_τ	< 0.1 eV	0	$\frac{1}{2}$	Stable	Same as above. Detected 2000.
<i>Quarks (stable in bound states)</i>					
tier1Up	2.2 MeV	$+\frac{2}{3}$	$\frac{1}{2}$	Stable*	Two ups + one down = proton. The most abundant massive particle.
tier1Down	4.7 MeV	$-\frac{1}{3}$	$\frac{1}{2}$	Stable*	One up + two downs = neutron. Stable inside nuclei.
<i>Force carrier (open branch)</i>					
tier1Photon	0	0	1	Stable	Electromagnetic radiation. Light. The open energy branch.

*Quarks are confined — never isolated. Stable within protons/neutrons.

Count: 4 fermions + 3 neutrinos + 1 photon = **8 persistent species** (plus their antiparticles).

These eight species (and their antiparticles) account for 100% of the visible universe. Every atom. Every photon. Every neutrino streaming through you right now. The universe is built from eight persistent things and nothing else.

3 Tier 2 — Unstable: the visitors

These particles are real — they’re detected, they leave tracks, they have measurable properties. But they DECAY. They don’t persist. They appear in high-energy collisions or cosmic ray showers and convert into Tier 1 particles within microseconds or less.

In SFT: Tier 2 fermions are closure modes that exist as valid self-consistent orbits but are **ENERGETICALLY UNSTABLE** — a lighter mode with the same quantum numbers exists, so they shed energy and transition

downward. Tier 2 bosons may be transient settlement events rather than persistent particles — the resonance of an interaction, not a thing that maintains itself.

Particle	Mass	Charge	Spin	Lifetime	SFT interpretation
<i>Unstable fermions — real closure modes that decay</i>					
tier2Muon	106 MeV	−1	$\frac{1}{2}$	$2.2 \mu\text{s}$	Higher closure mode. Decays to electron because lighter mode exists.
tier2Tau	1.78 GeV	−1	$\frac{1}{2}$	$2.9 \times 10^{-13} \text{ s}$	Heaviest charged lepton mode. Decays to lighter modes.
tier2Strange	95 MeV	$-\frac{1}{3}$	$\frac{1}{2}$	$\sim 10^{-10} \text{ s}^\dagger$	Heavier quark mode. Decays via mode transition.
tier2Charm	1.27 GeV	$+\frac{2}{3}$	$\frac{1}{2}$	$\sim 10^{-12} \text{ s}^\dagger$	Same.
tier2Bottom	4.18 GeV	$-\frac{1}{3}$	$\frac{1}{2}$	$\sim 10^{-12} \text{ s}^\dagger$	Same.
tier2Top	173 GeV	$+\frac{2}{3}$	$\frac{1}{2}$	$5 \times 10^{-25} \text{ s}$	Heaviest fermion. Decays before hadronizing.
<i>Massive gauge bosons — possibly transient settlement events</i>					
tier2W $^\pm$	80.4 GeV	± 1	1	$3 \times 10^{-25} \text{ s}$	Mode transition mediator. The weak interaction seen as a resonance.
tier2Z 0	91.2 GeV	0	1	$3 \times 10^{-25} \text{ s}$	Neutral mode transition. Same interpretation.
<i>Scalar boson</i>					
tier2Higgs	125 GeV	0	0	$1.6 \times 10^{-22} \text{ s}$	Settlement geometry excitation. The closure mechanism briefly ringing at its own frequency.
<i>Confined gauge boson</i>					
tier2Gluon	0	0	1	Never free	Settlement field between quarks. Never isolated. Seen only through jet patterns.

[†]Quark lifetimes are measured through their host hadrons.

Count: 6 unstable fermions + 3 bosons + gluon = **10 transient species**.

Every Tier 2 particle either decays into Tier 1 particles or is never observed in isolation. None of them constitute matter. None of them persist. They are the TRANSITIONS and RESONANCES of the system, not the system itself.

4 Tier 3 — Virtual or hypothetical: the ghosts

These have never been independently detected. They exist in the mathematics of QFT or in theoretical proposals. SFT claims none of them are physically real.

Particle	Status	SFT position
tier3Graviton	Hypothetical	Gravity is the compactness gradient $a = -c^2 \nabla C_A$. No carrier needed. Geometry doesn't need a messenger.
tier3Virtual photons	Math tool	Electromagnetic force is directional settlement compatibility. No exchange particle needed. The settlement geometry between charges creates the force directly.

Particle	Status	SFT position
tier3Virtual gluons	Math tool	Strong force is settlement at nuclear compactness. The “gluon exchange” is what settlement looks like in perturbation theory.
tier3Virtual W/Z	Math tool	Weak interaction is mode transition. The “exchange” is the interaction vertex.
tier3Dark matter particles	Hypothetical	Galaxy rotation curves explained by morphology-dependent λ from $\kappa = 3/4$. No invisible mass needed.
tier3Dark energy / Λ	Hypothetical	Modified Friedmann $H^2 = 2G\rho/(3\pi)$ from correct boundary accounting. No vacuum energy needed.
tier3SUSY partners	Hypothetical	No evidence at LHC. SFT does not require supersymmetry.
tier3Axion	Hypothetical	Strong CP solved by no-overwrite direction-blindness (candidate, not derived).
tier3Magnetic monopole	Hypothetical	Not required or predicted by SFT.

Count: 0 confirmed. All either mathematical infrastructure or undetected proposals.

5 The treasure map

SFT’s job is to derive every measurable property of the 8 Tier 1 species from one rule (no-overwrite) operating on one stuff (energy at c) in one spacetime (3+1 dimensions).

5.1 What SFT must explain for each Tier 1 particle

Property	Status	Current SFT position
Mass	Conditional	$L_{\text{eff}} = 3\pi - \frac{1}{2} \ln(25/13)$ gives hierarchy to 0.032%. Species map open.
Charge ($\pm 1, \pm \frac{2}{3}, \pm \frac{1}{3}, 0$)	Candidate	Circulation direction. Quantization and fractional charges open.
Spin ($\frac{1}{2}$)	Derived	4π spinor closure. $r_{\text{spin}} = \hbar/(2mc)$. $L = \hbar/2$.
Magnetic moment	Open	Not addressed. Anomalous moment measured to 13 digits.
$\alpha = 1/137$	Open	Directional-to-scalar settlement ratio. Mechanism identified, value not derived.
Color charge	Open	Possibly related to 3+1 slot occupancy structure. Not derived.
Quark confinement	Candidate	Settlement at nuclear compactness. Candidate, not derived.
Neutrino mass	Candidate	Precessing directional phase cancels charge. Reduced closure load 4π only. Order-of-magnitude candidate, not derived.
Neutrino oscillation	Candidate	No electromagnetic lock. Orbit wanders between modes. Not derived.
Left-handedness	Candidate	Precession-motion coupling selects one stable helicity. Not derived.
Stability / decay	Partial	Heavier modes decay because lighter modes with same quantum numbers exist. Specific rates open.

Property	Status	Current SFT position
Interactions	Open	Scattering, absorption, emission, annihilation, decay rules. Ledger interaction mechanics not formalized.

5.2 The assembly chain

The universe assembles from Tier 1 particles in a specific sequence, each step using only settlement mechanics:

Quarks → **Protons/Neutrons**. Three quarks bound by settlement at nuclear compactness. Two ups + one down = proton (stable). One up + two downs = neutron (stable in nuclei). Binding energy ~ 8 MeV per nucleon.

Protons + **Neutrons** → **Nuclei**. Nuclear fusion in the early universe (Big Bang nucleosynthesis) and in stars. Settlement binding against Coulomb repulsion. Creates helium, lithium, and eventually all elements through stellar nucleosynthesis.

Nuclei + **Electrons** → **Atoms**. Electromagnetic settlement binding. Electron orbits around nuclear settlement well. This IS recombination — the ledger crystallization from V13.2.

Atoms → **Molecules** → **Chemistry**. Shared settlement geometries between atoms. Chemical bonds. Molecular structure. The origin of complexity.

Atoms → **Gas** → **Stars**. Gravitational settlement accumulation from V13.1. Gas clouds collapse. Fusion ignites. Stars are gravitationally bound settlement structures.

Stars → **Heavy Elements** → **Planets**. Stellar nucleosynthesis creates elements beyond helium. Supernovae distribute them. Gravitational settlement forms rocky planets.

Planets → **Chemistry** → **Life**. Complex molecular settlement structures. Self-replicating chemistry. Biology. Eventually, a structure complex enough to look back and ask what particles are.

Every step uses only Tier 1 particles and settlement mechanics. No Tier 2 particles persist in the final structures. No Tier 3 particles are needed at any step. Eight species and the settlement geometry between them build everything.

6 What this means for the V13 program

The V13 program doesn't need to explain 61 particles. It needs to explain 8.

Four persistent fermions (electron, up, down, and their neutrino partners). One open boson (photon). Their masses. Their charges. Their spins. Their interactions. Their stability. And how settlement mechanics at different scales produces the forces that bind them into atoms, nuclei, stars, and galaxies.

The Tier 2 particles are CONSEQUENCES — transient states that appear when Tier 1 particles interact at high energy. The muon is the electron's heavier cousin. The top quark is the up quark's heaviest relative. The W boson is the weak interaction seen as a resonance. Explain the Tier 1 particles and their interactions, and the Tier 2 particles follow as excited states and transition resonances.

The Tier 3 particles are UNNECESSARY — mathematical tools that SFT replaces with settlement geometry, or hypothetical substances that SFT replaces with correct boundary accounting.

The treasure map in one sentence:

Derive the masses, charges, spins, and interaction rules of four fermions and the photon from energy at c with no-overwrite in 3+1 dimensions, and the universe assembles itself.